Roll No……..

MID SEMESTER EXAMINATION SEPTEMBER – 2025

Name of the Program: M.TECH

Name of the Course: Computer Aided Analysis of Mechanical System

Semester: 3rd

Paper Code: MME 152

Time: 90 Minutes

Maximum Marks: 50

Note:

(i) Answer ***all the questions*** by choosing ***any one of the sub questions***.
(ii) Each question carries 10 marks.

Q1	(10 marks)	
(a)	Classify different types of systems with suitable examples (mechanical, electrical, fluid, thermal)?	**CO1**
	OR	
(b)	Differentiate between static system design and dynamic system design with real-life applications?	**CO1**
Q2		
(a)	Discuss the importance of system classification in engineering problem solving?	**CO1**
	OR	
(b)	A system is represented by the stiffness matrix K = [[6, -2], [-2, 4]]. Find its eigenvalues?	**CO1**
Q3		
(a)	Explain different methods of vibration suppression used in engineering practice?	**CO2**
	OR	
(b)	What is coordinate coupling? Explain with a two-degree of freedom example?	**CO2**
Q4		
(a)	Differentiate between viscous damping and hysteretic damping with examples?	**CO2**
	OR	
(b)	A damped SDOF system has m = 5 kg, k = 2000 N/m, and damping coefficient c = 50 Ns/m. Calculate: (i) Natural frequency (undamped), (ii) Damped natural frequency, (iii) Damping ratio?	**CO2**
Q5		
(a)	Write short notes on vibration testing and its significance in mechanical system design?	**CO2**
	OR	
(b)	For a two-degree of freedom system with masses m1 = m2 = 1 kg and stiffness k1 = k2 = 1000 N/m, write the equations of motion and calculate the natural frequencies?	**CO2**